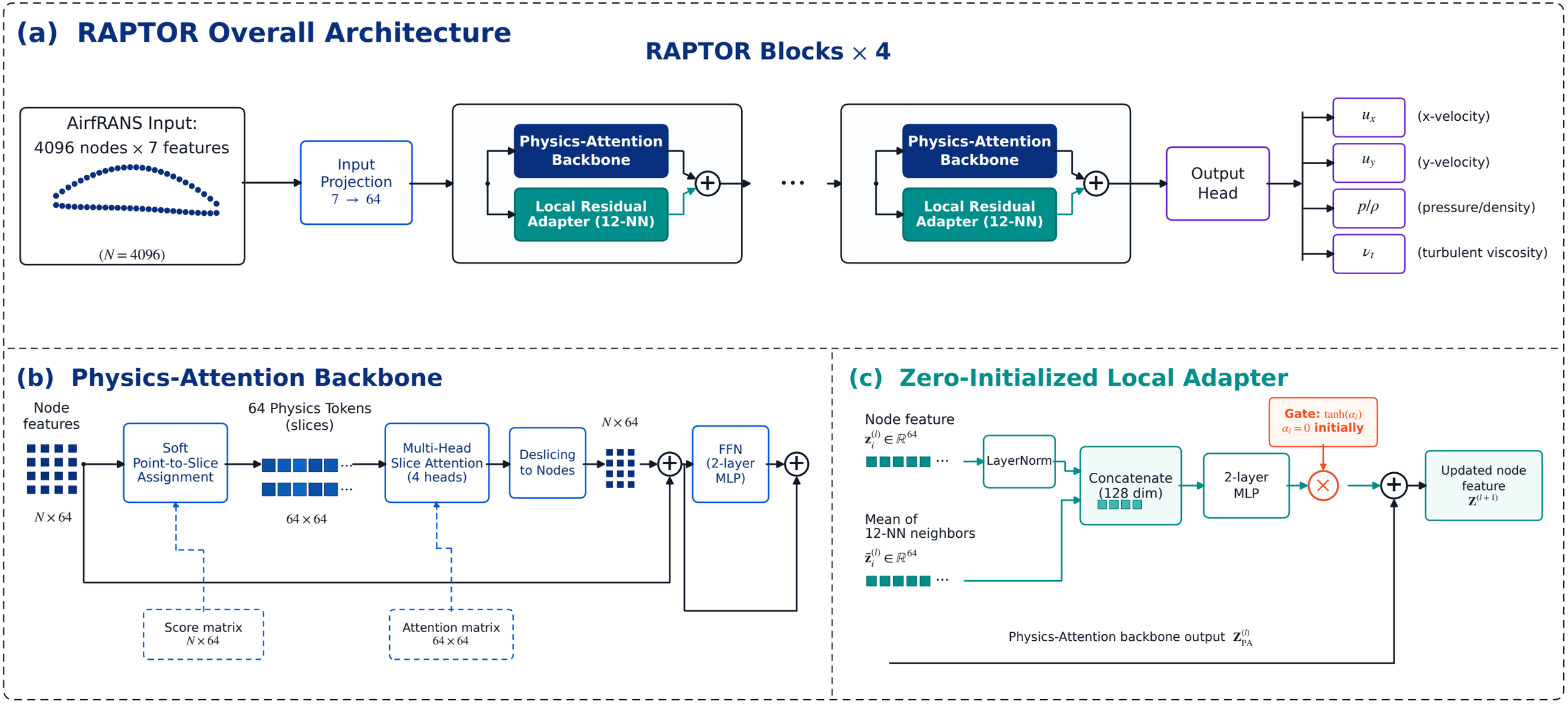
(a) RAPTOR Overall Architecture
RAPTOR Blocks × 4
AirfRANS Input:
4096 nodes × 7 features
(N = 4096)
Input Projection
7 → 64
Physics-Attention Backbone
Local Residual Adapter (12-NN)
Physics-Attention Backbone
Local Residual Adapter (12-NN)
Output Head
u_x (x-velocity)
u_y (y-velocity)
p/ρ (pressure/density)
ν_t (turbulent viscosity)
(b) Physics-Attention Backbone
Node features
N × 64
Soft Point-to-Slice Assignment
64 Physics Tokens (slices)
64 × 64
Multi-Head Slice Attention (4 heads)
Deslicing to Nodes
N × 64
FFN (2-layer MLP)
Score matrix
N × 64
Attention matrix
64 × 64
(c) Zero-Initialized Local Adapter
Node feature
$\mathbf{z}_i^{(l)} \in \mathbb{R}^{64}$
LayerNorm
Concatenate (128 dim)
2-layer MLP
Gate: tanh(α_l)
$\alpha_l = 0$ initially
Updated node feature
$\mathbf{Z}^{(l+1)}$
Mean of 12-NN neighbors
$\bar{\mathbf{z}}_i^{(l)} \in \mathbb{R}^{64}$
Physics-Attention backbone output $\mathbf{Z}_{\mathrm{PA}}^{(l)}$